

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, mean_squared_error
```

```
# Load the Breast Cancer Wisconsin dataset
```

```
data = load_breast_cancer()
```

```
X = data.data
```

```
y = data.target
```

```
print("Number of samples:", X.shape[0])
```

```
print("Number of features:", X.shape[1])
```

```
# Split the dataset into training and testing sets
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# Standardize the features
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
# MLE Logistic Regression
```

```
# penalty=None means no regularization
```

```
mle_model = LogisticRegression(
```

```
    penalty=None,
```

```
    max_iter=10000
```

```
)
```

```
mle_model.fit(X_train, y_train)
```

```
# Make predictions
```

```
mle_pred = mle_model.predict(X_test)
```

```
# Calculate accuracy
```

```
mle_accuracy = accuracy_score(y_test, mle_pred)
```

```
# MAP with L2 Prior
```

```

l2_model = LogisticRegression(
    penalty="l2",
    C=1,
    max_iter=10000
)

l2_model.fit(X_train, y_train)

# Make predictions
l2_pred = l2_model.predict(X_test)

# Calculate accuracy
l2_accuracy = accuracy_score(y_test, l2_pred)

# MAP with L1 Prior
l1_model = LogisticRegression(
    penalty="l1",
    solver="liblinear",
    C=1,
    max_iter=10000
)

l1_model.fit(X_train, y_train)

# Make predictions
l1_pred = l1_model.predict(X_test)

# Calculate accuracy
l1_accuracy = accuracy_score(y_test, l1_pred)

# Compare model performance
print("\nMODEL PERFORMANCE")

print("MLE Accuracy:", mle_accuracy)
print("MAP-L2 Accuracy:", l2_accuracy)
print("MAP-L1 Accuracy:", l1_accuracy)

# Compare parameter estimates
print("\nPARAMETER ESTIMATES")

print("MLE coefficients:")
print(mle_model.coef_)
print("\nMAP-L2 coefficients:")

```

```
print(l2_model.coef_)

print("\nMAP-L1 coefficients:")

print(l1_model.coef_)
```

```
Number of samples: 569
Number of features: 30

MODEL PERFORMANCE
MLE Accuracy: 0.9385964912280702
MAP-L2 Accuracy: 0.9736842105263158
MAP-L1 Accuracy: 0.9736842105263158

PARAMETER ESTIMATES
MLE coefficients:
[[ 9.43309636 -16.98638813  40.02092582  10.89014962   5.03344322
 274.94009818 -134.8381584 -266.18473606  40.99650503 -168.01794604
-259.63946217  35.91792789  115.35011389 -167.38940131 -79.54006031
-97.5741303   246.60809325 -70.05165414   95.95418597  131.93923412
-105.87135969 -91.92127648   73.01874488 -103.73668988   93.97496151
 29.28537604 -244.05971135  -8.90816016 -169.6548548   59.68383553]]

MAP-L2 coefficients:
[[-0.43190368 -0.38732553 -0.39343248 -0.46521006 -0.07166728  0.54016395
 -0.8014581  -1.11980408  0.23611852  0.07592093 -1.26817815  0.18887738
 -0.61058302 -0.9071857  -0.31330675  0.68249145  0.17527452 -0.3112999
 0.50042502  0.61622993 -0.87984024 -1.35060559 -0.58945273 -0.84184594
 -0.54416967  0.01611019 -0.94305313 -0.77821726 -1.20820031 -0.15741387]]

MAP-L1 coefficients:
[[ 0.          0.          0.          0.          0.          0.
 0.         -2.43008634  0.01930185  0.         -2.46930797  0.38212589
 0.          0.         -0.4338919  0.8877901  0.          0.
 0.46413608  0.28160816 -0.71476598 -1.85254945  0.         -3.16891716
 -0.16116272  0.         -1.29067207 -0.01750769 -1.03559545  0.         ]]
```